# Lecture 15

## Multivariate distributions

### The multinomial distribution (Discrete)

Context (Count multiple categories).

- Suppose we have items of $k$ different types ($k \geq 2$). [it generalizes 2 different types: Head/Tail; Failure/Success; etc.].

- Each type $i$ can occur with probability $p_i$, $1 \leq i \leq k$; $\sum_{i=1}^{k} p_i = 1$.

- These items characterize a general population

- Take a sample of size $n$ of such items, independently of each other and with replacement.

- We may want to know the probability that $x_1$ are of type 1,
  $x_2$ are of type 2,
  . . . . . .
  $x_k$ are of type $k$.

Definition 1. Let $X =: (X_1, \dots, X_k)$ be a random vector of counts (each component counts a type). Then, if, for $x_1 + x_2 + \dots + x_k = n$,

$$\mathbb{P}(X_1 = x_1, \dots, X_k = x_k) = \binom{n}{x_1, \dots, x_k} p_1^{x_1} \cdot \dots \cdot p_k^{x_k}$$

we say that $(X_1, \dots, X_k)$ has a multinomial distribution with parameters $n$ and $p =: (p_1, \dots, p_k)$.

- $\binom{n}{x_1, \dots, x_k} = \frac{n!}{x_1! \cdot \dots \cdot x_k!}$, the multinomial coefficient.

# Example 5.9.2 (text in book)

$X =: (X_1, X_2, X_3)$

$X_1$: # of people < 10 miles

$X_2$: # of people $\in [10, 50]$.

$X_3$: # of people > 50 miles

- Given: $X \sim \text{Multinomial}_3(n=20, (p_1 = 0.23, p_2 = 0.59, p_3 = 0.18))$

Want $\mathbb{P}(X_1 = 7, X_2 = 8, X_3 = 5) =$

$$= \frac{20!}{7!\,8!\,5!} (0.23)^7 \cdot (0.59)^8 \cdot (0.18)^5 = 0.0094.$$

Note If $X =: (X_1, \ldots, X_K)$ satisfies Definition 1, we write $X \sim \text{Multinomial}_K(n, p)$, with

$p =: (p_1, \ldots, p_K)$.

Example 5.9.3 (text in book).

$X_1$ = # of people with blood type A

$X_2$ = " " " " B

$X_3$ = " " " " AB

$X_4$ = " " " " O

Given: $(X_1, X_2, X_3, X_4) \sim Mult_4(n, (p_1, p_2, p_3, p_4))$

where $p_1 = 0.36$

$p_2 = 0.123$

$p_3 = 0.038$

$p_4 = 0.479$.

A sample of $n = 2$ people is drawn.
What is the probability that they have the same blood type?

Solution

Independence

$$\mathbb{P}(\text{Same blood type}) = \mathbb{P}(X_1=2, X_2=0, X_3=0, X_4=0)$$

$$+ \mathbb{P}(X_1=0, X_2=2, X_3=0, X_4=0) +$$

$$+ \mathbb{P}(X_1=0, X_2=0, X_3=2, X_4=0) +$$

$$+ \mathbb{P}(X_1=0, X_2=0, X_3=0, X_4=2)$$

$$= \binom{2}{2,0,0,0} \left[p_1^2 + p_2^2 + p_3^2 + p_4^2\right] = 0.376.$$

## Multinomial and Binomial

**Result 1** $(X_1, X_2) \sim \text{Multinomial}_2(n, (p_1, p_2))$,

then: $X_1 \sim \text{Bin}(n, p_1)$

$$X_2 = n - X_1.$$

Result 2 (All marginals of a multinom. are binomial).

If $(X_1, \ldots, X_K) \sim \text{Mult.}(n, (p_1, \ldots, p_K))$,

then $X_i \sim \text{Bin}(n, p_i)$, for

each $i = 1, \ldots, K$.

Result 3. $Y \doteq X_1 + X_2 + \ldots + X_\ell$, with

$\boxed{\ell < K}$, and $K > 2$ (in the context of Result 2). Then $Y \sim \text{Bin}(n, p_1 + \ldots + p_\ell)$.

(Note: The result is NOT true for $\ell = K$. General form in Corollary 5.9.2.)

**Result 4** If $X \sim \text{Multinomial}_k(n, (p_1, \ldots, p_k))$,

then [with $X =: (X_1, X_2, \ldots, X_k)$].

$$E(X_i) = n p_i, \text{ for each } i = 1, \ldots, k,$$

$$\text{Var}(X_i) = n p_i (1 - p_i), \text{ for each } i = 1, \ldots, k,$$

$$\text{cov}(X_i, X_j) = -n p_i p_j, \text{ for any } i \neq j.$$

**Example.**

- $p_1 = 10\%$ Freshmen $=: F$
- $p_2 = 14\%$ Sophomores $=: So$
- $p_3 = 40\%$ Juniors $=: J$
- $p_4 = 36\%$ Seniors $=: Se$.

- Select 20 students at random.
- What's the chance that there will be at least 5 $J$ & $Se$, combined.

Solution

- $X =: (F, So, J, Se) \sim \text{Multinomial}(20, (p_1, \ldots, p_4))$

  ↑ abuse of notation

- We don't distinguish between $J$ or $Se$, we only care about $J + Se$. By Result 3 above, $Y =: J + Se \sim \text{Binomial}(n = 20, p_3 + p_4)$

  $= \text{Bin}(n = 20, p = 0.76)$.

- To solve the problem, compute therefore

  $\mathbb{P}(Y \geq 5)$, for $Y \sim \text{Bin}(20, 0.76)$.

# The Bivariate Normal Distribution

- Say we want to propose a probability model (that is, give the expression of the distribution function) of $X_1$ = Cholesterol Level and $X_2$ = Body Mass Index (BMI). Clearly, they are linked, so we need to model them jointly, and define a bivariate distribution function for the vector $(X_1, X_2)$.

- In many applications, it is appropriate to use the Bivariate Normal (Gaussian) Distribution

**Definition** We say that $(X_1, X_2)$ follow a bivariate normal distribution with parameters:

$\mu_1, \mu_2$ (location)

$\sigma_1^2, \sigma_2^2$ (scale)

$\rho$ (correlation)

if the joint p.d.f. $f(x_1, x_2)$ of $(X_1, X_2)$ is:

$$(*) \quad f(x_1, x_2) = \frac{1}{2\pi(1-\rho^2)^{1/2}\sigma_1\cdot\sigma_2} \cdot \exp\left\{-\frac{1}{2(1-\rho^2)}\cdot\left[\left(\frac{x_1-\mu_1}{\sigma_1}\right)^2 + \left(\frac{x_2-\mu_2}{\sigma_2}\right)^2 - 2\rho\left(\frac{x_1-\mu_1}{\sigma_1}\right)\left(\frac{x_2-\mu_2}{\sigma_2}\right)\right]\right\}.$$

- The matrix $\Sigma = \begin{pmatrix} \sigma_1^2 & \rho\cdot\sigma_1\cdot\sigma_2 \\ \rho\cdot\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}$ is called the covariance matrix of $(X_1, X_2)$.
- The vector $\mu = (\mu_1, \mu_2)$ is called the mean (expected value) of $(X_1, X_2)$.

## Very important Properties

① $E(X_1) = \mu_1$, $E(X_2) = \mu_2$

② $Var(X_1) = \sigma_1^2$, $Var(X_2) = \sigma^2$

③ $\rho(X_1, X_2) = \rho$. ↑ Recall $\rho(X_1, X_2) = \frac{Cov(X_1, X_2)}{\sigma_1 \cdot \sigma_2}$

↑ the correlation

④ $X_1 \sim N(\mu_1, \sigma_1^2)$ and $X_2 \sim N(\mu_2, \sigma_2^2)$

**Theorem** (Very IMPORTANT) Assume that $(X_1, X_2) \sim MVN_2(\mu, \Sigma)$ ← Notation for the multivariate normal distribution in 2 dimensions)

$X_1$ and $X_2$ are independent if and only if

$\rho = 0$.

**Remark** Recall that in general lack of correlation is NOT equivalent with independence.

## Properties (Continued)

If $(X_1, X_2) \sim f(x_1, x_2)$ given by (*) at page 10, then

(1) $X_2 \mid X_1 = x_1$ has a (conditional) normal distribution.

(2) The mean and variances of this normal distribution depend on the value of $x_1$ and are given by the formulas below:

(3) $$E(X_2 \mid X_1 = x_1) = \mu_2 + \rho \cdot \sigma_2 \cdot \left(\frac{x_1 - \mu_1}{\sigma_1}\right)$$

(4) $$\text{var}(X_2 \mid X_1 = x_1) = (1 - \rho^2)\,\sigma_2^2.$$

- Similar results hold for $X_1 \mid X_2 = x_2$, see display (5.10.8) page 341 in book.

NOTE If you need to predict $X_2$ after observing $X = x_1$, in a way that minimizes the MSE, you need to use (3), and the associated MSE value will be (4)

**Result 10** If $(X_1, X_2) \sim f(x_1, x_2)$ given by ⊛,

**then**: $Y = a_1 X_1 + a_2 X_2 + b \sim N(\mu, \sigma^2)$,

**where** $\mu = a_1\mu_1 + a_2\mu_2 + b$

$$\sigma^2 = a_1^2\sigma_1^2 + a_2^2\sigma_2^2 + 2a_1a_2\rho\sigma_1\sigma_2.$$

**Example 1** $X_1$ = Results on test A
$X_2$ = Results on test B.

$(X_1, X_2) \sim f(x_1, x_2)$ with parameters:

$\mu_1 = 85$, $\mu_2 = 90$

$\sigma_1 = 10$; $\sigma_2 = 16$

$\rho = 0.8$.

**Calculate**: $\mathbb{P}(X_2 > 90 \mid X_1 = 80) = ?$

Use the result at page 12.

The distribution of $X_2$ given that $X_1 = 80$ is normal with parameters:

$$\mu = 90 + 0.8 \cdot 16 \cdot \left(\frac{80 - 85}{10}\right) = 83.6$$

and

$$\sigma^2 = (1 - 0.8^2)\, 16^2 = 92.16$$

Therefore, to simplify notation, let $X_2 \mid X_1 = 80 = Y$, and $Y \sim N(83.6, 92.16)$. Then

$$\mathbb{P}(Y > 90) = \mathbb{P}\left(\frac{Y - 83.6}{\sqrt{92.16}} > \frac{90 - 83.6}{\sqrt{92.16}}\right)$$

$$= \mathbb{P}\left(Z > \frac{2}{3}\right) \longrightarrow \text{use table!}$$

Example 2 In the context of Example 1, Calculate $\mathbb{P}(X_1 + X_2 > 200)$.

Solution Use Result 10 page 13 above.

Then $Y = X_1 + X_2 \sim N(\mu, \sigma^2)$, with

$$\mu = 1 \cdot 85 + 1 \cdot 90 + 0 = 175$$

$$\sigma^2 = 1 \cdot (10)^2 + 1 \cdot (16)^2 + 2 \cdot 1 \cdot 1 \cdot 0.8 \cdot 10 \cdot 16$$

$$= 612.$$

Therefore $\mathbb{P}(Y > 200) = \mathbb{P}\left(\frac{Y - 175}{\sqrt{612}} > \frac{200 - 175}{\sqrt{612}}\right)$.

$$= \mathbb{P}(Z > 1.0106) \longrightarrow \text{Use Table!}$$